Buck converter

330µ, 63 V, >1500 mA ripple current
Panasonic EEUFR1J331

4.7µ, 100V, X7S
TDK C3225X7S2A475K200AB

Solar Panel
Voltage: 16-50 V

SOLAR0A
SOLAR0B
D3 SB560
TP3 >TP_SIGNAL_NAME
R8 5m
R7 1M
D2 1N4148
TP4 >TP_SIGNAL_NAME
DCDC_IN
Q3 MOSFET-NCHANNELFDD8780
C1 330µ
C18 330µ
C2 4.7µ
C3 4.7µ
C30 4.7µ
PGND
R13 100k 0.1%
R14 680 0.1%
C16 100n
IN4148
IC3
+3V3
R15 200k
C15 100n
C14 100n
GND

Diode must survive
at least Vin+14V

+12V
C6 100n
D1 1N4148
IC4
TP1 >TP_SIGNAL_NAME
VDD
DR
DB
EN
RDT
VSS
HB
HO
HS
LO
LM5106
R22 t.b.d.
R1 t.b.d.
C5 100n
R21 t.b.d.
C7 t.b.d.
Q1 MOSFET-NCHANNELFDD8780
Q2 MOSFET-NCHANNELFDD8780
PGND
TP5 >TP_SIGNAL_NAME
22µ
L1
+12V
C4 820µ
C19 10µ
C24 10µ
LOADOB
LOADOA

8A: Coilcraft SER2918H-223
12A: Würth HCF 7443642200

820µ, 25 V
Panasonic EEUFR1EB21
10µF, X7R, 25V
TAIYO YUDEN TMK316B7106KL-TD

Battery management system

Balancing:
Rmin = 0.5*VCmax/50mA
Qloss < 0.1W

R6 4.7
R5 4.7
R4 4.7
R3 4.7
R2 4.7
C12 100n
C11 100n
C10 100n
C9 100n
C8 100n
X1E
X1D
X1C
X1B
X1A
BQ76920
NC
VC5
VC4
VC3
VC2
VC1
VC0
SRP
SRN
ALERT
BAT
REGSRC
REGOUT
CAP1
TS1
SCL
SDA
VSS
CHG
DSG
+12V
R17 100
+3V3_BAT
Boot signal
for bq76920
R19 1k
D7
TIMP0
C22 10µ
C21 1µ
C20 4.7µ
C17 1µ
C26 100n
R20 1k
R25 1M
ALERT pin drives interrupt in
case of new current reading
or battery error

BATOB
BAT0A
F1 10A
SMAJ15A
TP2 >TP_SIGNAL_NAME
C28 100n
C13 100n
C27 100n
R14 1k
R10 1k
R9 5m
R12 1M
R24 1M
C23 100n (t.b.d.)
R16 1M
Q6 BSSB4
D5 1N4148
R18 1M
Q5 MOSFET-NCHANNELFDD8780
Q4 MOSFET-NCHANNELFDD8780
Prevent ground offsets when FETs are off

Design: Martin Jäger
m.jaeger@posteo.de
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Arduino

+12V
C29 1µ
U1
IN
GND
EN
OUT
BP
V_REG_LDOSMD
+3V3
C25 4.7µ

VCC = 3.3V Output
Max Current Output: 150mA

FTDI0
GND
CTS
VCC
TXD
RXD
DTR
FTDI Basic

ICSP0
MISO
SCK
RESET
+5
MOSI
GND
AVR_SPI_PRG_6PTH

+3V3
C31 100n
DTR
R6 10k
RESET
U2
PC6(/RESET)
AVCC
VCC@1
VCC@2
AREF
PB6(XTAL1/TOSC1)
PB7(XTAL2/TOSC2)
PD0(RXD)
PD1(TXD)
PD2(INT0)
PD3(INT1)
PD4(XCK/T0)
PD5(T1)
PD6(AIN0)
PD7(AIN1)
PB0(ICP)
PB1(OC1A)
PB2(SS/OC1B)
PB3(MOSI/OC2)
PB4(MISO)
PB5(SCK)
AGND
GND@1
GND@2
ATMEGA168
C32 100n
Y1
RESONATORSMD
C33 100n
R45 2k2
R46 2k2
SDA
SCL

Status LED using PWM signal on D9
ON = charging
OFF = not charging
BLINK = something else

R25 1k
LED1
PWM9

Add RX/TX (D0 and D1)?

ANALOG
JP2
JP3
DIGITAL

Pin headers (2.54 mm pitch) break out the unused µC
pins and can be used for extensions like:
- display (e.g. EA-DOGS102)
- trimmer and push-buttons (e.g. to set PWM manually)
- communication (e.g. RS485 or CAN)

R47 330
LED2 Green

Design: Martin Jäger
m.jaeger@posteo.de
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